Gerardo Nieto

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Java Solutions Architect / Technical Leader / Sr FULL STACK Java Developer / Microservices Developer

Professional Summary

- 22+ years of experience in programming and **15+ years of experience building enterprise software applications** in different **industries like Medical, Financial, Government, Consultancy, and Retail**.
- Expertise in Software Development Life Cycle: **planning, defining requirements, design & prototyping, development, testing, deployment, operation & maintenance** using **Agile (Scrum), Waterfall, and Test-driven development (TDD).**
- **3+ years of experience with Microservices Stack *(Configuration management, Service discovery, Fault tolerance, Centralized logging, Centralized metrics, Distributed tracing, Scheduling, Deployment, Auto Scaling).***
- **3+ years of experience** with Cloud Technologies like ***Google Cloud Platform, Azure, Kubernetes, Docker, and API Gateways (Apigee, Layer7).***
- **Experience with Reactive Programming and Functional Programming.**
- Extensive expertise with:
  - Java **11 (*Type Inference*)**; Java **8 (*Streams and Lambda expressions*)**; Java **6 (*Collections*)**; Java **5 (*Generics*)**; Java **4** (***XML support***).
  - The J2EE platform like ***Servlets, Java Server Pages (JSP), Java Database Connectivity (JDBC), Java Persistence API (JPA), Enterprise JavaBean (EJB), Java Message Service API (JMS), Security, Java Transaction API (JTA), Java API for XML Web Services (JAX-WS), Java API for RESTful Web Services (JAX-RS)*.**
  - Java Frameworks like ***Spring (Reactive, Boot, MVC, Data, Security, Cloud, Apache Kafka, Web Services), Hibernate, Java Server Faces (JSF), Struts*.**
  - Web/Application Servers like ***Tomcat (v8, v9)*, *Oracle WebLogic, IBM WebSphere, Red Hat WildFly, and Oracle GlassFish*** for deploying and configuring web applications.
  - Web Design like ***Angular, HTML, HTML Document Object Model (DOM), XHTML, Extensible Markup Language (XML), Extensible Stylesheet Language (XSLT), Cascading Style Sheets (CSS), JavaScript, Asynchronous JavaScript and XML (AJAX), jQuery, JavaScript Object Notation (JSON)*.**
  - Relational Database Management System (RDBMS) like ***Oracle 12g, SQL Server, MySQL, PostgreSQL*** using queries, **PL-SQL**, stored procedures, functions, and views.
  - NoSQL databases like ***MongoDB, Google Firestore, and Azure Cosmos*.**
  - Messaging tools like ***Apache Kafka.***
  - Enterprise Service Bus (ESB) like ***TIBCO*.**
  - Testing frameworks like ***JUnit, Mockito, and REST-assured*.**
  - Development tools like ***Eclipse, IntelliJ, NetBeans, SQL Developer, Postman, SOAP-UI, NoSQLBooster, Robomongo, and TOAD*.**
  - Workflow and Decision Automation Platform tools like ***Camunda***.
  - Project management tools like ***Maven*.**
  - Text editors like ***Sublime Text and Notepad++*.**
- Proficient in the integration of tooling for **CI/CD** like **Nexus Repository Manager, Sonar, Looper, and Jenkins**
- Work with version control systems like **SVN, Git, and GitHub.**
- Project Tracking Software like **Jira**
- Strong Core Java with a strong understanding of principles of Object-Oriented paradigm: **Encapsulation, Abstraction, Inheritance, Polymorphism; Multi-threading, Exception handling, Garbage Collector, Design patterns (Data Access Object, Data Transfer Objects, Abstract Factory, Service Data Objects, Business Object, Singleton, Interfaces, Business Delegate), Unified Modeling Language (UML)**.
- Learn **Python** to work with analysis of Big Data.
- Learn **Scala** to improve Functional Programming concepts.
- Job experience like **Sysadmin** on **UNIX**, and **Linux** platforms, developed shell scripts, and configured **CRON** jobs.

Technical Skill	
Programming Paradigms	**Reactive and Functional Programming, Object-Oriented, Procedural.**
Front End Technologies	**Angular (15), HTML5, DOM, XML, XSLT, JavaScript, AJAX, jQuery, JSON, CSS, JSF, JSP, Struts**

Back End Technologies	**Java, J2EE, JDBC, JPA, JMS, JTA, JAX-WS, JAX-RS, Spring, Spring Boot, Spring Data, Spring Security, Spring Cloud, Hibernate, Apache Avro, ESB Tibco, Unix bash scripting, Python, Scala.**
Testing	**JUnit, Mockito, REST-assured, Sonar, JMeter, VisualVM**
Messaging	**Apache Beam, Apache Kafka, Apache ActiveMQ**
Decision Automation	**Camunda**
Management	**Maven, Gradle**
Cloud Technologies	**Spring, Google Clous Platform, Azure, Apigee, Layer7, Dataflow, Airflow**
CI/CD and DevOps	**Docker, Kubernetes, Jenkins, Looper**
Databases	**Oracle 12g, SQL Server, MySQL, PostgreSQL, MongoDB, Cosmos, Cassandra**
Reporting	**Jasper**
Application Servers	**Tomcat, WildFly, WebLogic, WebSphere, GlassFish**
Other	**Jira, Visio**
Versioning	**git, svn**
Documentation	**UML**
Methodology	**Agile SCRUM, Waterfall, Test-driven development**
O.S.	**Red Hat Enterprise Linux, Fedora, Windows, Mac OS X**

Education

National Polytechnic Institute Mexico City 2001- 2006
Bachelor's in Computer Systems Engineering

Work Experience

Client: Family Savings and Loan Cooperative, **Remote**
Company: Personal project **Jan 2025 – current**
Role: Java Solutions Architect

Description: Savings and Loan Cooperative is moving its financial system from a monolithic legacy to cloud architecture. As a solution architect, I am responsible for designing and implementing the business logic oriented to a native cloud application, selecting appropriate technologies that are scalable, secure, and maintainable over time, and aligned with the organization's strategic objectives.

Responsibilities as Architect:

- Work closely with stakeholders to understand the requirements, constraints, and goals.
- Translate functional and non-functional requirements into a technical solution.
- Evaluate technologies to determine the best fit based on business requirements.
- Design the solution, deliver technical specifications and architecture diagrams, and define software, hardware, and network components.

Environment: AWS, Google Cloud Platform, Architectural Patterns.

Client: Equifax, **Atlanta, GA**
Company: Active Alliance / Mindtree **Oct 2022 – Apr 2025**
Role: Technical Leader / Full Stack Developer

Description: Equifax consolidates information from multiple partners to create financial products for its clients. Due to its expansion, I aim to develop the necessary plugins to clean, import, and integrate the partners' data into Equifax's infrastructure, moving mainframe-based business logic to the Cloud environment, adhering to its standards and architecture. Besides, it has an area called Workforce Solutions to develop tools like "I-9 HQ" designed to help Human Resources teams and new hires save their I-9 Form information and stay in contact during the hiring process.

Responsibilities as Technical Leader:

- Understand Equifax´s business logic and the technological core to make it comprehensible to the developers; this includes understanding Equifax's tools (ETL process) developed under the Google Cloud Platform.
- Be responsible for leading a group of developers and testers in creating the Equifax configuration files and Java files necessary to load source information into GCP. The technology used here is Apache Avro and internal Equifax APIs developed under the Google Cloud Platform.
- Document and explain all internal processes to make the logic used in Equifax understandable to programmers.
- Improve the skills of developers and QA testers.

Responsibilities as Full Stack Developer:

- Understand I-9 Web Portal business logic developed under Angular, Equifax APIs, and Google Cloud Platform.
- Create and modify Angular components to respond to new business requirements in the web portal, such as uploading files, making HTTP calls, and updating UI.
- Create Angular test scenarios related to Angular components.
- Use Jenkins CI/CD pipelines to deploy Angular code into environments.

Environment: Java 11, Google Cloud Platform (Kubernetes Engine, Cloud Storage [Buckets], BigQuery, Pub/Sub, Dataflow, Airflow, Dataprep, Firestore Database, BigTable Database), Apache Avro, Angular, Internal Equifax APIs.

Client: Adobe, **Houston TX**
Company: Active Alliance / Nextgentech **Jan 2022 – Sep 2022**
Role: Software Developer Java 3

Description: Adobe maintains hundreds of millions of records related to User Experience from its clients. Due to storage being generally an expensive resource, my objective was to create a first-approach statistics of its Big Data to save storage and see correlations between data.

Responsibilities:

- Developing **Rest services** with **Java 18** and **Spring Boot** as **microservices** and improving internal Java projects dedicated to testing the User Experience pipelines.
- To Learn the meaning of the data.
- To create a first-approach solution to visualize the correlations between the data graphically using **Python, Pandas** (a Python library for working with data sets)**, Matplotlib** (a low-level graph plotting Python library)**, Java,** and **Gradle**.

Environment: Java 18, Python 3, Gradle.

Client: EY (Ernst & Young), **Houston TX**
Company: Active Alliance / Pyramid **Jul 2021 – Oct 2021**
Role: Reactive Microservices developer

Description: EY, one of the Big Four accounting firms, is developing ***reactive*** **event-driven microservices** to perform internal tasks to create executive reports for their clients. As a reactive microservices developer **using Spring Reactive** technology, my mission was to validate the performance of the microservices through load tests. Update the business decision rules implemented in Camunda. Create unit test cases to validate business logic behavior. The tech stack includes **Azure (Portal and DevOps), Camunda, Apache Kafka, Azure Cassandra, and Azure Databricks**.

Responsibilities:

- Create test cases to validate **reactive programming and functional programming** methods.
- Modify **Camunda's** Business Process Model and Notation (**BPMN**) and Decision Model and Notation (**DMN**) files.
- Give support load testing phases, including creating internal statistics and error analysis.
- Update microservices building knowledge using **Spring Reactive Flux and Mono** objects.
- The **first approach to Python** as a language preferred to create Azure Databricks notebooks.
- Use **Azure DevOps** as a code repository, CI/CD, and planning tool.
- Use **Azure Portal** as PaaS

Environment: IntelliJ, Java 11, Spring Reactive, Spring Boot, Kafka, Azure Cassandra, Postman, Git, Maven, Junit.

Client: Walmart USA, Mexico City
Company: Softtek **Mar 2020 – Jun 2021**
Role: Architect / Technical Lead / Microservices developer

Description: Walmart USA, the biggest retail company globally, is migrating its Mainframe business core to the Cloud in an effort called MF2C. As the team leader, my team and I were assigned to create the architecture and develop a complete cloud-based solution to migrate the mainframe-based flow using new and legacy **microservices, Kafka, NoSQL,** and **PaaS** as the technical stack.

Responsibilities:
- Understand the core business logic at high (Walmart), medium (Accounts Payable), and low (Store Receivings) levels.
- Analyze the issues related to using two different programming paradigms and their contrasting conceptions to organize the data: procedural/relational vs. object-oriented/non-relational.
- Led a team of 4 developers, responsible for task allocation, mentoring, best code practices, issue resolutions, pair programming, and architecture design.
- Generate the Architecture using **microservices**, queues, and transaction logic to represent the core logic implemented on the mainframe side.
- Migrate mainframe logic through reading mainframe code to create a high-level specification to be implemented in Java.
- Design the Representational State Transfer (**REST**) modules: *data collection, data enrichment, store-receivings core, confirmations, and reports*.
- Validate the data migration from a Relational Database to a NoSQL database performed by another team.
- Developing **Rest services** with **Java 8** and **Spring Boot** as **microservices**, these web services were deployed in Walmart Cloud Native Platform (**WCNP**) infrastructure.
- Develop the messages and payloads to interchange data between the services.
- Implement **Kafka** consumers and producers to use the asynchronous messaging and make a fault-tolerant solution.
- Use **Git** as a code repository.
- Use **Maven** as a management tool.
- Use **Azure Cosmos** as a NoSQL database.
- Use **Looper** as a tool for CI/CD
- Use **JUnit**, **REST-assured** as the testing framework.

Environment: IntelliJ, Java 8, Spring Boot, Kafka, DB2, Azure Cosmos, Postman, NoSQLBooster, Robomongo, Git, Maven, Jenkins, JUnit, REST-assured.

Client: Softtek Microservices Academy, Mexico City
Company: Softtek **Oct 2019 – Feb 2020**
Role: Lead Instructor of Microservices Academy

Description: Softtek is convinced that developing the technical skills of new talent is the key to achieving the organization's objectives. Therefore, my objective as Microservices Lead Instructor was, in 4 months, 80% of participants must have the necessary knowledge to develop business applications oriented to the Microservices Architecture.

Responsibilities:
- Led a team of 2 instructors to create the material and labs to teach and develop the technical knowledge of 20 beginner programmers in MS Architecture.
- Topics:
  - **Git** as code repository
  - **Maven** as a project management tool
  - **Domain-Driven Design**
  - Unified Modeling Language (**UML**)
  - **Java basics (abstraction, encapsulation, inheritance, polymorphism, threads)**
  - **Java 8 (Lambdas, Streams)**

- Hypertext Markup Language (**HTML**)
- **Servlets** & Java Server Pages (**JSP**)
- Java Database Connectivity (**JDBC**)
- **Persistence** and Object-Relational Mapping (**ORM**)
- **Object-Oriented Patterns Design** (***Creational***: Abstract Factory, Singleton. ***Structural***: Composite, Decorator, Facade)
- **Spring MVC**
- **Junit** as a Unit Testing tool
- **Spring Security**
- Microservices fundamentals
  - **Configuration management** like Netflix Archaius**,**
  - **Service discovery** like Eureka**,**
  - **Fault tolerance** (Circuit breaker) like Hystrix**,**
  - **Centralized logging** like Elasticsearch, Logstash, Kibana**,**
  - **Centralized metrics,**
  - **Distributed tracing** like Splunk**,**
  - **Scheduling & deployment** like Jenkins pipelines**,**
  - **Auto-scaling** like Docker and Kubernetes**.**
- Microservices Design Patterns
  - **Aggregator,**
  - **API Gateway,**
  - **Asynchronous Messaging,**
  - **Command and Query Responsibility Segregation (CQRS),**
  - **Saga,**
  - **Circuit Breaker.**
- **Spring Boot**
- **REST** services
- **Spring Cloud**
- **Swagger**
- **Postman**
- Message Brokers
  - **ApacheMQ**
- **OAuth 2.0**
- **API Managers (apigee)**
- **Repository Managers** (Nexus)
- Containers & Images (**Docker**)
- **Continuous Integration** (Testing, Build Images, Push Artifacts)
- **Continuous Deployment (Jenkins, Kubernetes)**
- Developed the exercises to evaluate the participants.

Environment: Java 8, JSP, servlets, Tomcat, Spring Boot, Spring Cloud, Swagger, Postman, ApacheMQ, Git, OAuth 2.0, Maven, UML, Bitbucket, Hibernate, Apigee, Nexus, Heroku Cloud Application Platform, Jenkins, Docker, IntelliJ.

Client: Brunswick, Frisco TX
Company: Softtek **Mar 2019 – Sep 2019**
Role: Microservices Developer

Description: Brunswick is trying to move its Monolith Architecture to a new **Microservices** Architecture. My objective was to migrate a Java backend project to a microservices project using the best practices to demonstrate the benefits of Microservices Architecture.

Responsibilities:
- Responsible for analyzing the legacy Java source code.
- Migrate the legacy source code to a standard convention over configuration (**Maven**) of Microservices Architecture.
- Create the java project to include **Spring Boot**, **Spring Data**, **JPA**, Azure Insights connections, and the **Oracle** database.

- Developed and executed Unit Test plans using **JUnit** and **REST-assured,** ensuring that results were documented and reviewed with the client.
- Create a blueprint document to explain the use of the Java tech stack.

Environment: Java 1.6-1.8, J2EE, Spring Boot, Spring Data, JPA, Hibernate 4, Maven, JUnit, REST-assured, IntelliJ, Oracle 12g, Scrum.

Client: MoneyGram, Frisco TX
Company: Softtek **Oct 2018 – Feb 2019**
Role: Technical Leader

Description: MoneyGram has a big part of its infrastructure in split modules of different technologies and vendors, maintaining communication through an Enterprise Service Bus. The objective was to migrate distributed backend services to microservices. Update deployment process to CD/CI approach and improve user experience by developing new functionality using its tech stack.

Responsibilities:
- Analyze technical information (diagrams, Entity-Relationship schemas, tools) about the customer's business model to be explained to all team members.
- Understanding legacy process (Simple Object Access Protocol messages using **ESB Tibco**) and tech stack (IBM **Master Data Management**, IBM **Operational Decision Manager**) to propose new solutions at the microservices level using Java as tech stack.
- Use technical specifications to apply them to new in-house developments.
- Understand legacy Java code to migrate to the new microservices structure.
- Understand functional testing used by the QA area to help to create more testing scenarios using **SoapUI**.
- Identify development best practices and code standards applied in Moneygram to be implemented by our team members.

Environment: Spring Boot, Java 8, Layer 7, Tibco (ESB), MDM (IBM), ODM (IBM), Maven, SoapUI, Oracle 12g, Couchbase (NoSQL), Bitbucket, Jenkins, Eclipse, SOAP (Big Services), Scrum and Kanban boards, Jira, Windows.

Client: Social Development Ministry, Mexico City
Company: Grupo Altavista **Apr 2016 – Sep 2018**
Role: Middleware Engineer

Description: The Social Development Ministry has web applications to accomplish its mission based on the National Development Plan. This infrastructure includes the first approach to distributed systems.

Responsibilities:
- Administrate services in **JBoss** Enterprise Application Platform (EAP) v5 & v6.
- Administrate Linux boxes installed on Red Hat Enterprise Linux (**RHEL**) v6 & v7.
- Responsible for developing, maintaining, performing, and monitoring the middleware environment using JBoss and RHEL, creating Linux boxes, configuring tools inside Linux boxes, creating users, and modifying permission access.
- Debug, resolve, and optimize code's developers to be deployed on JBoss reading code and using **VisualVM**.
- Identify, research, and resolve defects within a given application.
- Identify JVM thread dumps and memory leaks using IBM Thread and Monitor Dump Analyzer for Java (**TDMA**).
- Guide QA Team to implement test scenarios like availability, performance, code quality, response time, and usability using tools such as **JMeter** and **SonarQube**.
- Use **bash scripting** to create an automated deployment process. First contact with DevOps philosophy: **Jenkins, Red Hat OpenShift, Red Hat Ansible, Kubernetes**.
- Linux system administration.
- Manage Red Hat Enterprise Virtualization (**RHEV**) console.
- Use Apache Subversion (**SVN**) as the control versioning tool.
- Know customer's business.

Environment: RHEL 6 & 7, JBoss EAP 5 & 6, Java 6 & 7, JavaMail API, Unix bash scripting, JSF, Seam Framework, JPA, Hibernate, Maven, Oracle 12g, Jenkins, Artifactory, JMeter, SonarQube (Checkstyle, PMD, FindBugs), IBM Thread and Monitor Dump Analyzer for Java (TDMA), VisualVM, Eclipse, SVN, Fedora Distribution.

Client: Supremo Consejo de México, Mexico City
Company: Artiffex Tekton **Mar 2016 – Mar 2016**
Role: Java Architect

Description: The executive board needs control over the participants' inscription to the next congress. Therefore, my objective was to create a complete portal to register the congressman and offer administrative information.

Responsibilities:
- Create the complete website.
- Generate the Entity-Relationship (12 tables) diagram.
- Database mapping (**Hibernate**) and implementation using public **MySQL** web hosting.
- Design web application (**HTML**, **JSP**) architecture using security and MVC patterns with **Spring Security** and **Spring MVC**.
- Developing and implementing the system using public web hosting.
- Generate executive information using the Java API for Microsoft Documents (**Apache POI**).
- Use **GitHub** as the control versioning tool.
- Use **Maven** as software project management.
- Technical support.

Environment: Java 8, Spring MVC 4.0, Spring Security, Hibernate, JSP, Apache POI, HTML, jQuery, AJAX, JSON, JavaScript, CSS, Maven, MySQL, MySQL Workbench, Tomcat, Eclipse, Git, GitHub, Mac OS X.

Client: Lithomat, Mexico CIty
Company: Artiffex Tekton **May 2014 – Feb 2016**
Role: Java Architect

Description: The CEO of the print shop needs to automate its entire business process with Information Technology. My objective was to create an Enterprise Resource Planning (ERP) to allow the management of the print shop. Generate executive information on stocks, sales, clients' accounts, accounts receivable and accounts payable, human resources, and maintenance. Generate reports in Excel and PDF.

Responsibilities:
- Interview with stakeholders to get functional and non-functional requirements and describe business logic flow using Unified Modeling Language (**UML**) diagrams.
- Understanding patterns of the production process.
- Generate and implement an Entity Relationship diagram (64 tables) to accomplish the functional requirements using **MySQL** and **MySQL Workbench**.
- Design web application architecture using Java technology and MVC pattern and security (**Spring MVC**, **Spring Security**, **Hibernate**, **JSP**, **HTML**, raw **JavaScript**, **CSS**).
- Implement the system into a local **Mac OS X** server.
- Use **GitHub** as the code repository.
- Generate executive reports used by the CEO through a secured module using **JasperReports**.
- Technical support.

Environment: Java 6, Spring MVC 4.0, Spring Security, Hibernate, JSP, Servlets, JasperReports, HTML, jQuery, AJAX, JSON, JavaScript, CSS, Maven, MySQL, MySQL Workbench, Tomcat, Eclipse, Git, GitHub, Mac OS X.

Client: Low-cost airline Interjet, Mexico City
Company: Artiffex Tekton **May 2012— Apr 2014**
Role: Sr Java Developer

Description: The marketing office needs to generate brand presence in social networks and obtain its clients' possible consumer patterns. Therefore, my objective was to create multiple web applications to get the airline's quality feedback and create numerous web applications to reward the client's loyalty.

Responsibilities:

- Design, develop, and implement multiple Entity relationship diagrams.
- Design, develop, and implement multiple web application architectures.
- Generate manual executive information from marketing quizzes using in-house queries.
- Technical support.

Environment: Java 6, Spring MVC 4.0, Spring JDBC, JSF, HTML, AJAX, JavaScript, CSS, MySQL, Tomcat, JDeveloper, Mac OS X.

Client: Department of the Treasury, Mexico City
Company: Department of the Treasury **Oct 2009 – Apr 2012**
Role: Web Java Developer

Description: Federal Treasury daily admits hundreds of payment declarations sent to the Federal Tax Administration service. My objective was to analyze and improve code from the legacy Swing application. Develop a new web application using Spring MVC technology. Unify the business logic of separated modules related to creating and retrieving payment declarations into a unique web application. Generate further crossed information from payment declarations. Maintain levels of information security and encryption required by the law.

Responsibilities:

- Understanding Federal Treasury business logic.
- Debug the legacy **Java Swing** applications to get the high-level requirements.
- Understanding steps are done to encrypt data in-house.
- Develop improvements in database schema adding tables, indexes, and triggers.
- Developing web application using the J2EE framework: **JSP** and the first contact with **Spring MVC**.
- Maintenance of legacy applications until production lives.
- Technical leader.
- Technical support.

Environment: Java 6, Java Swing, Spring MVC 3.0, JSP, HTML, AJAX, JavaScript, CSS, MySQL, BEA WebLogic Application Server, Oracle, JDeveloper, Windows.

Client: Argos Afore, Mexico City
Company: Pagoss Consultory **Nov 2008 – Sep 2009**
Role: Java Developer

Description: Argos Afore is a financial institution responsible for pension savings management. My objective was to give continuity to the business by developing improvements related to the view: JSP, Servlets. Create new reports using XSLT in the XML data. Improve database queries.

Responsibilities:

- Learn mastering concepts such as Extensible Markup Language (**XML**) and Extensible Stylesheet Language Transformations (**XSLT**).
- Use the Java API from XML Processing (**JAXP**) to manage XML documents using Document Object Model (**DOM**), Simple API for XML (**SAX**), and XSLT.
- Learn **Apache Struts** (struts.xml) framework to implement the Model-View-Controller pattern.
- Use the JavaServer Pages Tag Library (**JSTL**) to encapsulate functionality related to reporting financial data expressed in XML format.
- Understanding business concepts to understand the Entity-Relationship diagram and create complex queries to represent new reports.

- Working with Stored Procedures.
- Manual testing.

Environment: Java 5, Struts, JSP, JSTL, XML, XSLT, JAXP, DOM, SAX, HTML, JavaScript, CSS, Oracle, Windows.

Client: Elektra, Mexico City
Company: Elektra **May 2008 – Oct 2008**
Role: Java Developer

Description: Elektra is a Mexican retail store with many stores in the country. My objective was to help senior programmers make maintenance in-house web applications used by the Human Resources department.

Responsibilities:
- Learning mastering concepts as Hypertext Transfer Protocol (**HTTP**), **Servlets** (*javax.servlet, and javax.servlet.http packages*), JavaServer Pages (**JSP**), Hypertext Markup Language (**HTML**), Cascading Style Sheets (**CSS**), **JavaScript**.
- Learning Servlet Lifecycle: *init(), service(), doGet(), doPost(), doHead(), doPut(), doDelete(), destroy().*
- First contact with servlets/JSP in a **web application** (*web.xml*) that does not contain the Model-View-Controller pattern.
- Use filters (*web.xml + javax.servlet.Filter*) to validate authentication and authorization of request for resources.
- Maintenance of the JSP/servlets used at the web project related to evaluating employees.
- Use Java Database Connection (**JDBC**) to get data from a database.
- Technical support.

Environment: Java 4, HTTP, Servlets, JSP, JDBC, Windows.

Client: Medica Sur Hospital, Mexico City
Company: Medica Sur Hospital **Jul 2006 – Apr 2008**
Role: Programmer

Description: The Hospital administration is very complex. There are many data subdomains related to each other to give a complete health service required by the patient. My objective was to help senior programmers extend the modules implemented in the in-house Electronic Health Record (EHR). This EHR was built in OpenEdge Advanced Business Language (Progress 4GL).

Responsibilities:
- Mastering the concepts of Relational Database.
- Create a complete module to help schedule patients at the Radiology Department. This module includes the development of a new **Entity-Relationship diagram** of 32 tables.
- Generate a new Graphical User Interface (**GUI**) to manage the Scheduling module.
- Improve queries to consume data already populated in other tables unrelated to the ER diagram.
- Technical support.

Environment: Progress 4GL, Windows.